The Baseline Problem: Silent Failures in Construction Supply Chains

The baseline problem addressed by the FlowSense architecture is the phenomenon of "silent failures" within construction supply chains. The document identifies that once materials are ordered, visibility into their status effectively disappears until a critical issue arises.

This lack of transparency manifests in several ways that negatively impact project execution:

- Unnoticed Disruptions: Issues such as delayed shipments or missed fabrication windows occur without immediate detection.
- Cascading Schedule Impacts: Because delays are not caught in time, they cause a ripple effect that disrupts subsequent downstream activities. The document explicitly notes that delay cascades are the primary cost driver on construction projects.
- Costly Recoveries: By the time a project manager becomes aware of a material issue, it is typically too late to implement a recovery plan without incurring additional costs.

Three Unanswered Questions:

The core of the problem is summarized by three fundamental questions that traditionally remain unanswered on construction projects until a failure occurs:

- Current Location: Where exactly the material is located at any given moment, synthesizing data across all potential sources.
- Realistic Arrival Time: When the material will actually be delivered, factoring in the realities of fabrication timelines and logistics, rather than relying solely on vendor promises.
- Downstream Impact and Triage: If a material is delayed, what other project components will break as a result, and what specific actions should be prioritized to mitigate the damage.

The FlowSense platform is designed specifically to resolve this baseline problem by providing real-time answers to these three questions.